The Routing Paradox

Conserved information fixes what is preserved — but not how it routes

The setup, proven on a 5-node graph ($\beta_1 = 3$ independent cycles):

$B\ J = s$ conservation fixes the source boundary —
$J = J_0 + Z\ a$ — but every cycle coefficient a gives the same conservation. Routing is free.

two ways to resolve the ambiguity

GATE B · THE CLOSED DOOR

Self-Explaining System

The system selects its own routing from structure alone. No observer needed.

DOES NOT CLOSE

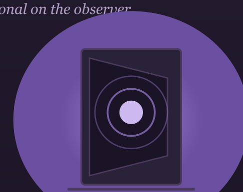
Why it fails — and fails irreducibly:

- The cycle space Z is the exact kernel of the boundary operator.
- Conservation, topology, Genesis anchoring — none fix a.
- Hodge selection works only after a metric W is supplied — and the network does not canonically supply W.
- *Fails at 5 nodes → not compute, not scale: it is algebra.*

GATE A · THE DOOR THAT OPENS

Interaction Selects Routing

A response operator R resolves the cycles. Closure is conditional on the observer.



CLOSES — CONDITIONALLY

The closure condition:

$\mathrm{rank}(R\ Z) = \dim(Z)$

When the observer resolves every hidden cycle, routing becomes identifiable — verified exact (error ≈ 3×10^{-16}).

A rank-deficient observer leaves the rest as gauge.

WHAT GATE A'S OBSERVER MUST BE

the same condition, read two ways

INSIDE THE SIMULATION

modest, local, achievable

- Any interaction coupling differently to different cycles — an interior subsystem.
- Need not span all cycles at once: local R resolves local cycles, leaves rest gauge.
- Finite, dissipative measurement is fine — the model has no thermodynamic cost.

→ Here, the observer is just: interaction.

IF APPLIED TO THE UNIVERSE (if it even does)

extraordinary, global, demanding

- To collapse universe-scale routing globally, R must resolve b_1 cycles — functionally infinite.
- A single global selector would sit outside the geometry it measures — not made of it.
- Resolve infinite loops without adding heat or decoherence to the system it observes.

→ Local interaction, OR something far stranger.

The universe column is conditional on the framework applying at all — a bridge not yet built. Held open, not asserted.

Where this lands

A conserved structure leaves a selection ambiguity it cannot resolve from within.
Either it closes on itself *(ruled out — even at 5 nodes)* or interaction selects the actual state.

This is the shape of the measurement problem.

Structure underdetermines; what selects is contested. Physics has not closed its Gate B either.

SPECIAL CLASS · ACKNOWLEDGED, NOT YET CATEGORIZED

A proven structural theorem + an empirically-validated recoverability law (batteries, hearts) — two real arms, not yet shown to be one mechanism.

Claim boundary: a finite-graph source-current identifiability theorem. No spacetime, ADM, or GR is derived. The resonance with physics is structural kinship, not proof.